

```
import nltk
import spacy
```

TOKENIZING:

medical_text = """ The patient was diagnosed with diabetes mellitus and hypertension. He is taking medications including metformin and antihypertensive drugs. Doctors are monitoring blood glucose levels and heart conditions. """

```
import nltk
nltk.download('punkt')
nltk.download('punkt_tab') # Added to download the re

medical_text = """
The patient was diagnosed with diabetes mellitus and
He is taking medications including metformin and anti
Doctors are monitoring blood glucose levels and heart
"""

from nltk.tokenize import sent_tokenize, word_tokeniz

sentences_nltk = sent_tokenize(medical_text)
words_nltk = word_tokenize(medical_text)

print(sentences_nltk)
print(words_nltk)
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
['\nThe patient was diagnosed with diabetes mellitus and hypertension.', 'He is
['The', 'patient', 'was', 'diagnosed', 'with', 'diabetes', 'mellitus', 'and', 'h
```

```
print("Sentences:")
print(sentences_nltk)

print("\nWords:")
print(words_nltk)
```

```
, 'monitoring', 'blood', 'glucose', 'levels', 'and', 'heart', 'conditions', '.']
```

Tokenization using spaCy:

```
import spacy

nlp = spacy.load("en_core_web_sm")
doc = nlp(medical_text)

sentences_spacy = [sent.text for sent in doc.sents]
words_spacy = [token.text for token in doc]

print(sentences_spacy)
print(words_spacy)
```

```
['\nThe patient was diagnosed with diabetes mellitus and hypertension.\n', 'He i\n', 'The', 'patient', 'was', 'diagnosed', 'with', 'diabetes', 'mellitus', 'an
```

Stemming NLTK:

```
from nltk.stem import PorterStemmer

stemmer = PorterStemmer()

stemmed_words = [stemmer.stem(word) for word in words_nltk if word.isalpha()]
print(stemmed_words)
```

```
['the', 'patient', 'wa', 'diagnos', 'with', 'diabet', 'mellitu', 'and', 'hyperte
```

```
['the', 'patient', 'wa', 'diagnos', 'with', 'diabet', 'mellitu',  
'and', 'hypertens', 'he', 'is', 'take', 'medic', 'includ', ...]
```

```
['the',  
'patient',  
'wa',  
'diagnos',  
'with',  
'diabet',  
'mellitu',  
'and',  
'hypertens',  
'he',  
'is',  
'take',  
'medic',  
'includ',  
Ellipsis]
```

Lemmatization spaCy

```
lemmatized_words = [token.lemma_ for token in doc if token.is_alpha]
print(lemmatized_words)
```

```
['the', 'patient', 'be', 'diagnose', 'with', 'diabete', 'mellitus', 'and', 'hype
```

```
['the', 'patient', 'be', 'diagnose', 'with', 'diabetes', 'mellitus',
 'and', 'hypertension', 'he', 'be', 'take', 'medication',
 'include', 'metformin', 'and', 'antihypertensive', 'drug', ...]
```

```
['the',
 'patient',
 'be',
 'diagnose',
 'with',
 'diabetes',
 'mellitus',
 'and',
 'hypertension',
 'he',
 'be',
 'take',
 'medication',
 'include',
 'metformin',
 'and',
 'antihypertensive',
 'drug',
 Ellipsis]
```



Close

conclusion:
in healthcare NLP,
lemmatization is preferred
over stemming because it
preserves the semantic
meaning of medical
terminology, ensures
clinical accuracy, and
improves performance in
tasks such as diagnosis
prediction, medical record
analysis, and decision
support systems.

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